

National Technical University of Athens

School of Electrical and
Computer Engineering (ECE)

10th semester

Control Systems of Electrical Machines

Simulation Assignment 2: Induction Motor Modelling and Direct-on-Line Starting

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1 Induction Motor Rated Quantities

Machine Parameters

A typical three-phase induction motor rated at 5 kW is selected for the modelling. The machine parameters used are presented in Table (1):

Quantity	Symbol	Value (SI)	Value (p.u.)
Rated power	P_n	5 kW	1
Rated voltage	V_n	400 V	1.732
Rated current	I_n	7.22 A	1
Frequency	f	50 Hz	1
Pole pairs	n_p	2	–
Synchronous speed	n_s	1500 rpm	1
Rated speed	n_n	1450 rpm	0.97
Slip	s	0.033	0.033
Stator resistance	R_s	1 Ω	0.0312
Rotor resistance	R_R	0.45 Ω	0.0141
Magnetizing inductance	L_M	0.20 H	1.96
Leakage inductance	L_σ	0.025 H	0.245
Rated torque	T_n	32.92 Nm	1.034
Moment of inertia	J	0.05 kg·m^2	–
Damping coefficient	b	0.001 Nms	–

Table 1: Parameters of the IM machine

Per-Unit Bases

The bases used for conversion to the per-unit system are given below. The machine phase voltage and current are used as bases, taking into account the normalization factor K. Since the quantities are normalized with respect to their RMS values, $K = \frac{1}{\sqrt{2}}$ is selected.

$$S_{base} = 5 \text{ kVA} \tag{1}$$

$$V_{base} = \sqrt{2}K\frac{400}{\sqrt{3}} = 230.94 \text{ V} \tag{2}$$

$$I_{base} = \sqrt{2}KI_n = 7.22 \text{ A} \tag{3}$$

$$Z_{base} = \frac{V_{base}}{I_{base}} = 32 \ \Omega \tag{4}$$

$$\omega_{base} = 2\pi f = 314.16 \text{ rad/s} \tag{5}$$

$$T_{base} = \frac{P_{base}}{\omega_{base} \cdot p} = 31.83 \text{ Nm} \tag{6}$$

$$\psi_{base} = \frac{V_{base}}{\omega_{base}} = 1.0396 \text{ Wb} \tag{7}$$

Justification of the Selected Parameters

The per-unit values are selected according to the typical parameters of an induction motor rated below 100 kW:

$$L_M = 1.5 - 3, \quad L_\sigma = 0.15 - 0.3, \quad R_s, R_R = 0.01 - 0.1 \tag{8}$$

Specifically, the magnetizing inductance $L_M = 1.96$ p.u. lies near the middle of the range $1.5 - 3$, while the leakage inductance $L_\sigma = 0.245$ p.u. lies within the range $0.15 - 0.3$. The resistances R_s and R_R are selected as approximately 0.03 and 0.015 p.u., respectively, corresponding to realistic copper losses for a machine of this rating. Four poles ($p = 2$ pole pairs) give a synchronous speed of 1500 rpm at 50 Hz, while the rated speed is slightly lower because of slip ($s \approx 3\%$), as expected. Finally, the mechanical parameters J and b are selected to represent the typical mechanical behaviour of a machine of this size.

2 Dynamic Model in the $\alpha\beta$ Reference Frame

The equations of the machine's inverse-Γ equivalent circuit were used to implement the dynamic induction motor model in the stationary $\alpha\beta$ reference frame:

$$L_\sigma \frac{di_s^s}{dt} = v_s^s - R_s i_s^s - \frac{d\psi_R^s}{dt} \tag{9}$$

$$\frac{d\psi_R^s}{dt} = R_R i_s^s - \left(\frac{R_R}{L_M} - j\omega_r\right)\psi_R^s \tag{10}$$

Applying the Laplace transform with zero initial conditions:

$$sL_\sigma I_s^s(s) = V_s^s(s) - R_s I_s^s(s) - s\Psi_R^s(s) \tag{11}$$

$$s\Psi_R^s(s) = R_R I_s^s(s) - \left(\frac{R_R}{L_M} - j\omega_r\right)\Psi_R^s(s) \tag{12}$$

Substituting Equation (12) into Equation (11) gives:

$$sL_\sigma I_s^s(s) = V_s^s(s) - R_s I_s^s(s) - R_R I_s^s(s) + \left(\frac{R_R}{L_M} - j\omega_r\right)\Psi_R^s(s) \Rightarrow \tag{13}$$

$$I_s^s(s) = \frac{1}{sL_\sigma + R_s + R_R}\left[V_s^s(s) + \left(\frac{R_R}{L_M} - j\omega_r\right)\Psi_R^s(s)\right] \tag{14}$$

Let:

$$I_s^s(s) = I_s^\alpha(s) + jI_s^\beta(s), \quad \Psi_R^s(s) = \Psi_R^\alpha(s) + j\Psi_R^\beta(s), \quad V_s^s(s) = V_s^\alpha(s) + jV_s^\beta(s) \tag{15}$$

Separating the real and imaginary parts and solving for the currents gives:

$$I_s^\alpha(s) = \frac{1}{sL_\sigma + R_s + R_R}\left[V_s^\alpha(s) + \frac{R_R}{L_M}\Psi_R^\alpha(s) + \omega_r\Psi_R^\beta(s)\right] \tag{16}$$

$$I_s^\beta(s) = \frac{1}{sL_\sigma + R_s + R_R}\left[V_s^\beta(s) + \frac{R_R}{L_M}\Psi_R^\beta(s) - \omega_r\Psi_R^\alpha(s)\right] \tag{17}$$

Similarly, solving Equations (16) and (17) for the flux linkages gives:

$$\Psi_R^\alpha(s) = \frac{1}{s + \frac{R_R}{L_M}}\left[R_R I_s^\alpha(s) - \omega_r\Psi_R^\beta(s)\right] \tag{18}$$

$$\Psi_R^\beta(s) = \frac{1}{s + \frac{R_R}{L_M}}\left[R_R I_s^\beta(s) + \omega_r\Psi_R^\alpha(s)\right] \tag{19}$$

The electromagnetic torque is given by:

$$\tau_e = \frac{3n_p}{2K^2}\mathrm{Im}\{\Psi_R^{s*} I_s^s\} = \frac{3n_p}{2K^2}\mathrm{Im}\{(\Psi_R^a - j\Psi_R^\beta)(I_s^\alpha + jI_s^\beta)\} = \frac{3n_p}{2K^2}\left(\Psi_R^\alpha I_s^\beta - \Psi_R^\beta I_s^\alpha\right) \tag{20}$$

Finally, the rotational speed is given by:

$$J\frac{d\omega_r}{dt} = \tau_e - \tau_L - b\omega_r \tag{21}$$

Figure (1) below shows the model implemented in Simulink.

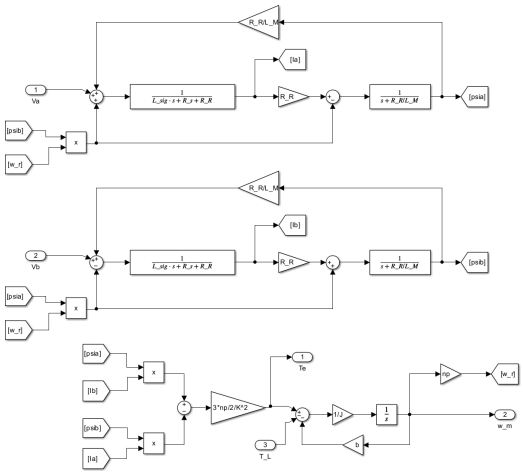


Figure 1: IM Motor model

3 Steady-State Induction Motor Model

In this question, the steady-state induction motor model was implemented in `MATLAB`. Specifically, the electromagnetic torque and stator current were calculated as functions of the rotor angular speed. The stator-current magnitude and electromagnetic torque are given by the following equations:

$$|i_s| = \frac{s}{s_p X_\sigma}\frac{|V_s|}{\sqrt{\left(\frac{s}{s_p}\right)^2 + 1}} \tag{22}$$

$$\tau_e = \frac{3n_p}{2K^2}\frac{s}{\omega_1 R_R}\frac{|V_s|^2}{\left(sX_\sigma/R_R\right)^2 + 1} \tag{23}$$

MATLAB Implementation

The main part of the MATLAB code used is shown below:

```
%% 3. Torque & Current vs rotor speed
% Supply voltage (phase RMS)
V_s = V_n / sqrt(3);

% Electrical frequency
w1 = 2*pi*f;

% Rotor angular speed
w_vec = linspace(0, w1, 1000);
w_r_pu = w_vec / w1;

```

```
% Slip
s = (w1 - w_vec) ./ w1;

X_sig = w1 * L_sig;
s_p = R_R / X_sig;

% Current
is_mag = s .* V_s ./ (s_p .* X_sig .* sqrt((s./s_p).^2 + 1));

% Torque
Te = (3*np/(2*K^2)) * (s/(w1*R_R)) .* (V_s^2 ./ (((s .* X_sig / R_R).^2) + 1));
```

Simulation Results

Figures (2) and (3) show the steady-state current and torque, respectively. These waveforms, and all subsequent waveforms, are normalized using the parameters and bases in Table (1).

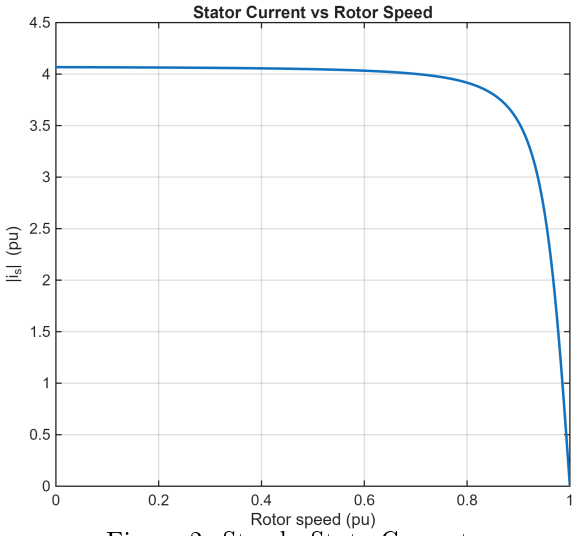


Figure 2: Steady-State Current

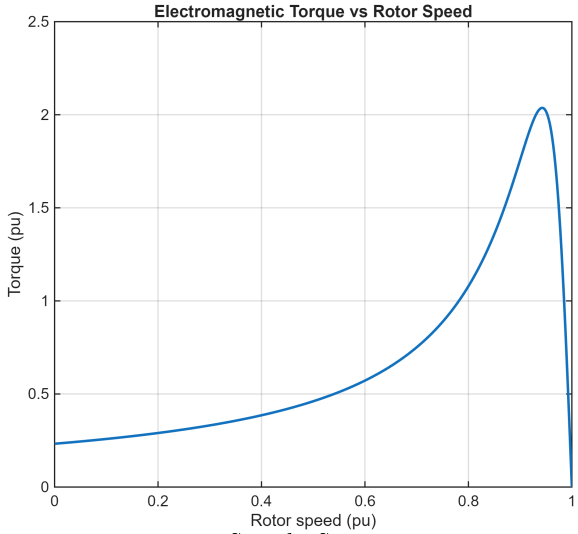


Figure 3: Steady-State Torque

For angular speeds below 0.85 *p.u.*, the current remains high, close to 4 *p.u.*, which is a typical starting-current value for an induction motor connected to the grid. When the rotor angular speed ω_r approaches the synchronous speed of 1 *p.u.*, the stator current becomes zero because the generated back electromotive force matches the rated voltage of the supply grid in both magnitude and phase.

The starting torque, i.e. the torque at low rotational speeds, is approximately 0.3 *p.u.*, as expected for this connection. It reaches a maximum of 2 *p.u.* at a particular slip, the pull-out slip: $s_p = R_R/X_\sigma = 0.0573$. Furthermore, for slip values greater than the pull-out slip, i.e. angular speeds greater than $1 - 0.0573 = 0.9427$ *p.u.*, rotational speed and torque appear to have a linear relationship, with torque becoming zero as synchronous speed is approached.

4 Direct-on-Line Motor Starting

In this question, direct-on-line starting of the induction motor was simulated using the dynamic model implemented in Question (2). The machine starts from rest and is supplied directly by the three-phase grid voltage, with zero load torque. Figure (4) shows the implemented model, and Figure (5) shows the Clarke-transformation subsystem.

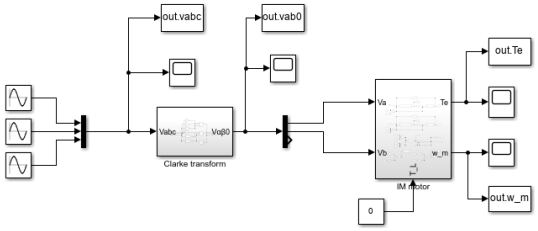


Figure 4: IM Motor model

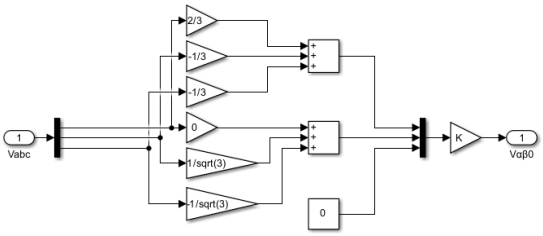


Figure 5: Clarke transform sub-system

Clarke Transformation

The Clarke transformation converts three-phase abc quantities into a stationary orthogonal $\alpha\beta$ coordinate system, according to Equation (24):

$$\begin{bmatrix} v_\alpha(t) \\ v_\beta(t) \end{bmatrix} = K \begin{bmatrix} \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ 0 & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} \end{bmatrix} \begin{bmatrix} v_a(t) \\ v_b(t) \\ v_c(t) \end{bmatrix} \tag{24}$$

Figures (6) and (7) below show the three-phase quantities before transformation and the corresponding $\alpha\beta$ components after applying the Clarke transformation.

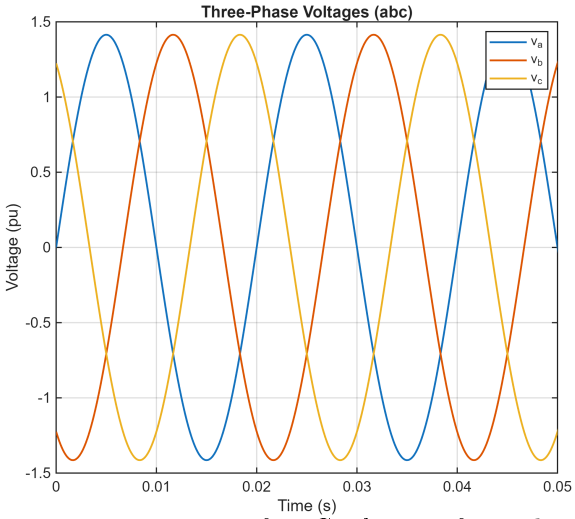


Figure 6: Voltage before Clarke transform abc

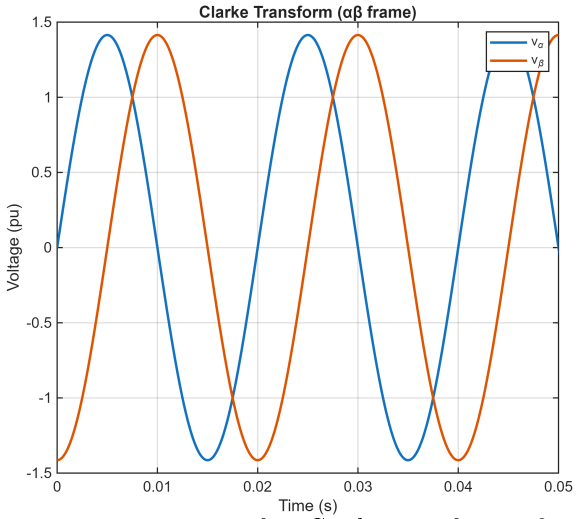


Figure 7: Voltage after Clarke transform $\alpha\beta$

Simulation Results

Figures (8)–(11) show the rotor speed, electromagnetic torque, rotor-flux magnitude and stator current during motor starting.

A pronounced transient occurs during starting, with large oscillations in torque, current and flux. In particular, rotor speed exhibits smaller oscillations than the other quantities. As speed increases, slip decreases and the system settles into steady-state operation.

In contrast, torque exhibits strong oscillations exceeding its rated value at the beginning of the transient, reaching its rated value when the slip is very small, as expected. After this peak, torque becomes negative to decelerate the machine slightly and then approaches zero because the mechanical damping load is very small.

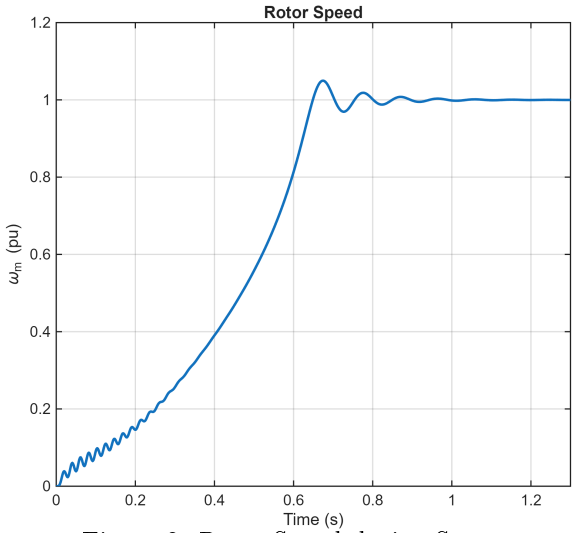


Figure 8: Rotor Speed during Start

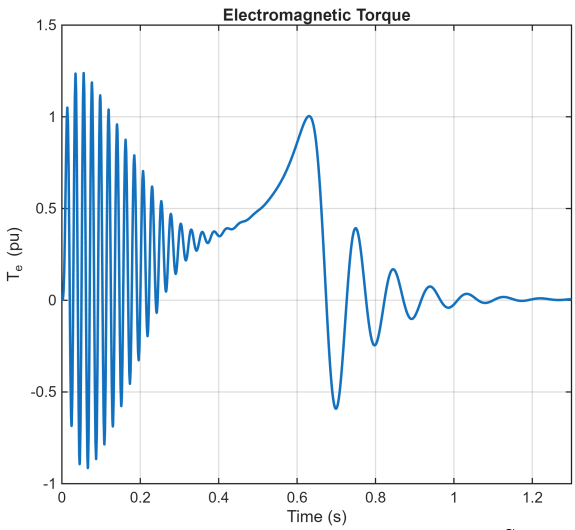


Figure 9: Electromagnetic Torque during Start

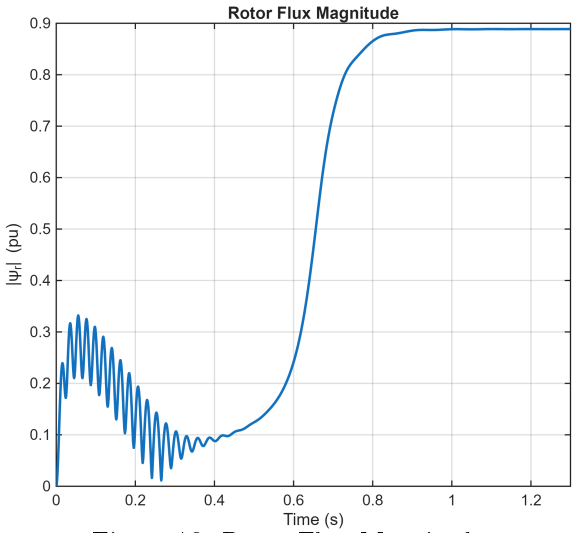


Figure 10: Rotor Flux Magnitude

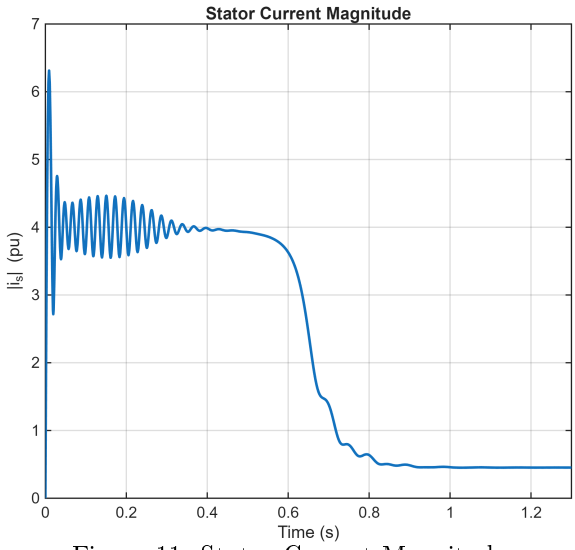


Figure 11: Stator Current Magnitude

Rotor flux also oscillates during the transient and then increases at a rate similar to rotational speed until it reaches a constant steady-state value. Note that this steady-state value is not 1 $p.u.$, since the rated rotor flux of the induction motor is approximated by:

$$\Psi_{N,pu}^{R} = \sqrt{\frac{1 - L_{\sigma,n}^{2}}{1 + 2L_{\sigma}/L_{M}}} = 0.8671 \tag{25}$$

Finally, during the oscillations, the stator current exceeds 6 $p.u.$, noticeably higher than the steady-state model result of 4 $p.u.$. It then remains at a high value of 4 $p.u.$ until the motor approaches its rated operating speed. At that point, the current does not become zero but retains a low value of approximately 0.4 $p.u.$ because of the magnetizing current.

5 Starting with the Steady-State Model

In this question, induction motor starting was simulated using the steady-state model implemented in Question (3). The main assumption is that the electrical quantities are in steady state at every time step, i.e. the flux is considered constant, and depend exclusively on the instantaneous rotor angular speed. Zero load torque is also assumed in this case.

For this reason, electromagnetic torque and stator current are calculated at every instant by linear interpolation from the previously calculated characteristics $T_e(\omega_r)$ and $I_s(\omega_r)$. Rotor speed is then obtained from the mechanical equation:

$$J\frac{d\omega_r}{dt} = \tau_e - T_L - b\omega_r \tag{26}$$

MATLAB Implementation

The main part of the MATLAB code used in combination with the preceding calculations is shown below.

```
% Simulation parameters
dt = 1e-5;
t_end = 0.8;
t = 0:dt:t_end;

% Initialization
T_L = 0;
w_r = 0;
w_m = 0;
w_r_vec = zeros(size(t));
Te_vec = zeros(size(t));
is_vec = zeros(size(t));

for k = 1:length(t)

    % Interpolation
    Te_interp = interp1(w_vec, Te, w_r, 'linear');
    is_interp = interp1(w_vec, is_mag, w_r, 'linear');

    % Mechanical dynamics
    dw = (Te_interp - T_L - b*w_m) / J;
    w_m = w_m + dw * dt;

    w_r = w_m * np;

    % Store values
    w_r_vec(k) = w_r;
    Te_vec(k) = Te_interp;
    is_vec(k) = is_interp;
end
```

Simulation Results

Figures (12)–(14) show rotor speed, electromagnetic torque and stator current during starting, as obtained from the steady-state model, together with the dynamic-model results obtained previously.

Comparing the steady-state and dynamic-model results reveals the following differences.

First, the torque and current oscillations of the dynamic model are entirely absent from the steady-state model. In particular, the maximum torque in this case is approximately 2 *p.u.*, compared with the approximately rated peak torque of the previous model. The maximum starting current is approximately 4 *p.u.*, versus more than 6 *p.u.* in the full model. Finally, the steady-state model appears to reach equilibrium, i.e. rated motor speed, approximately 0.2 s earlier.

All these differences arise because a very large magnetizing inductance, $L_M >> L_\sigma$, was assumed when deriving the torque and current equations (22) and (23), so the flux is considered constant. Consequently, the slow flux variations, whose time constant is comparable to that of the mechanical system, are not taken into account.

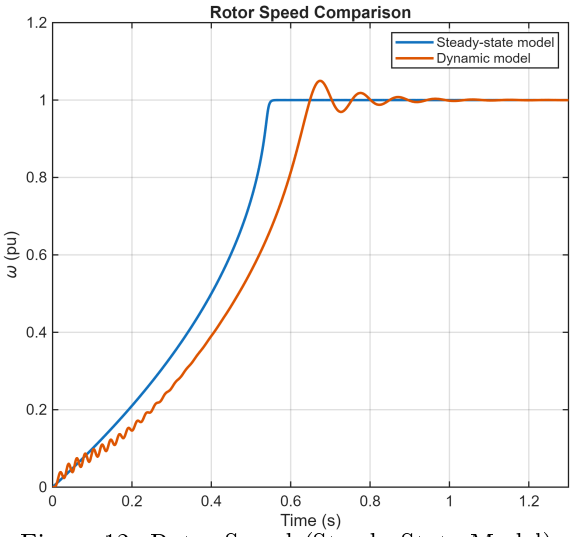


Figure 12: Rotor Speed (Steady-State Model)

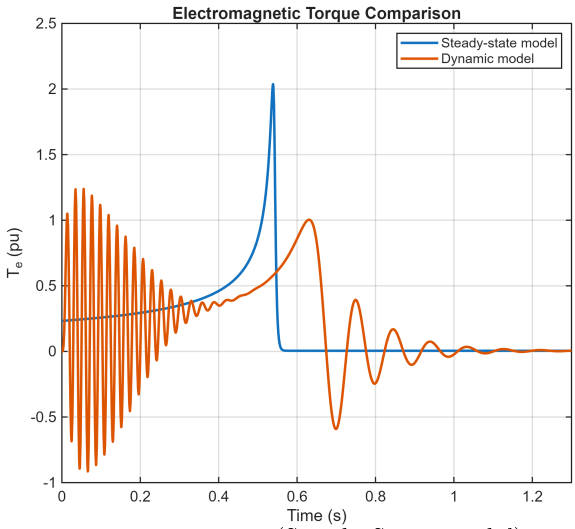


Figure 13: Torque (Steady-State Model)

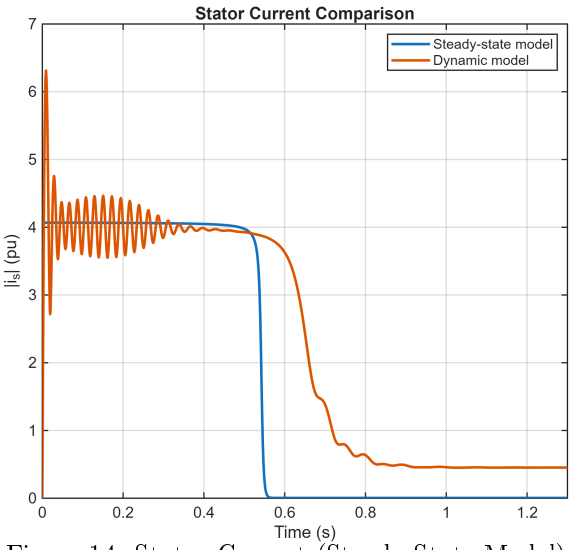


Figure 14: Stator Current (Steady-State Model)

6 Motor Starting with a Speed-Dependent Load

In this question, direct-on-line starting of the induction motor was simulated again using the dynamic model of Question (4). Unlike the previous case, the applied load torque depends on rotor speed. Two cases are examined: (a) a linear dependence and (b) a quadratic dependence on speed.

The proportionality constants were selected so that the steady-state load torque takes its rated value at rated rotational speed. The rated torque of the induction motor is given by:

$$T_{e,N} = \frac{3n_p}{2K^2}\frac{V_{base}I_{base}}{\omega_{base}}\frac{1}{1+2L_\sigma/L_M}\sqrt{(1-L_{\sigma,n}^2)\left[\left(1+\frac{L_\sigma}{L_M}\right)^2-\frac{1}{L_{M,n}^2}\right]} = 24.76 \text{ Nm} \tag{27}$$

$$T_{e,N} = 0.778 \; p.u. \tag{28}$$

Comparison of Loads Proportional to Speed and Speed Squared

In the first case, the load is proportional to rotational speed:

$$T_L = \frac{T_{e,N}}{\omega_{m,N}} \omega_m \tag{29}$$

In the second case, the load is proportional to the square of rotational speed:

$$T_L = \frac{T_{e,N}}{\omega_{m,N}^2} \omega_m^2 \tag{30}$$

Figures (15)–(18) compare rotor speed, electromagnetic torque, rotor-flux magnitude and stator-current magnitude for the two load cases.

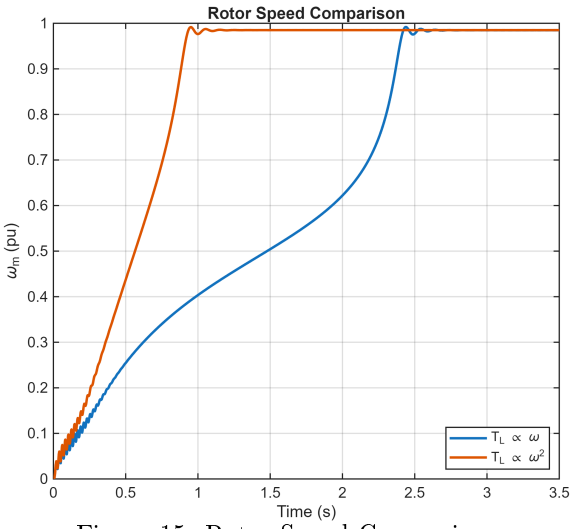


Figure 15: Rotor Speed Comparison

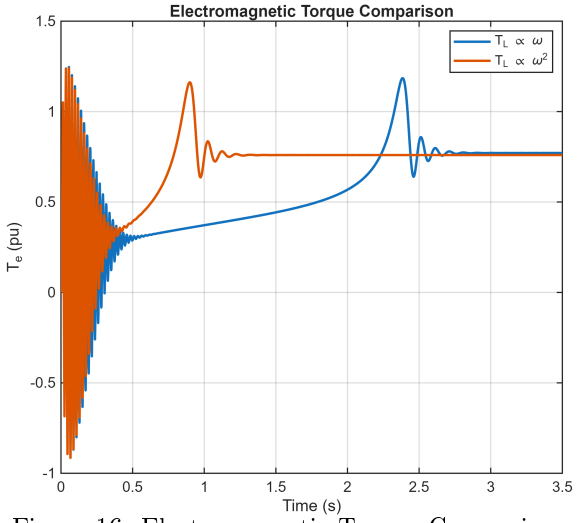


Figure 16: Electromagnetic Torque Comparison

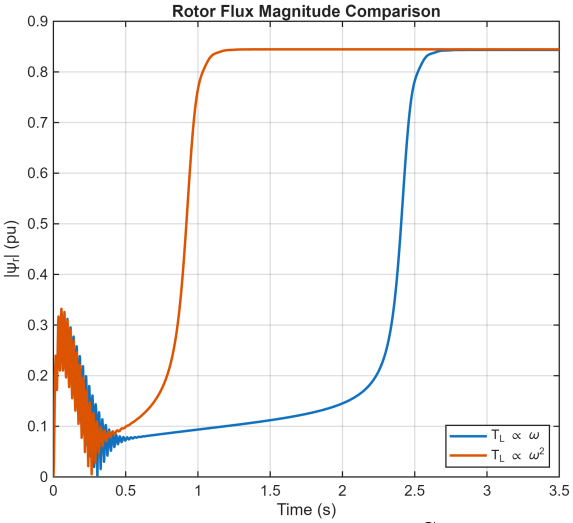


Figure 17: Rotor Flux Magnitude Comparison

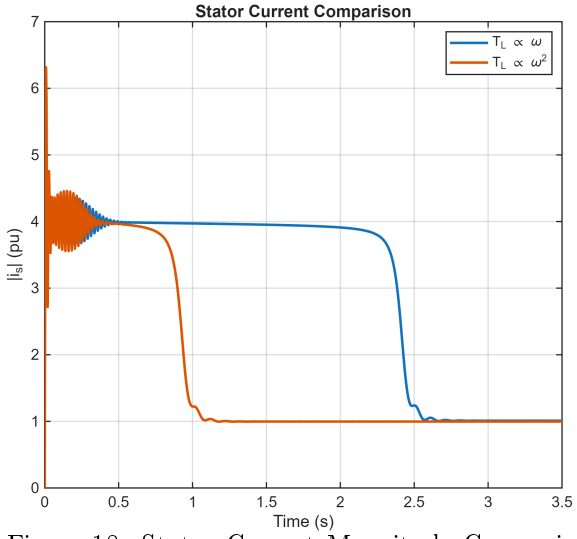


Figure 18: Stator Current Magnitude Comparison

The comparison plots show that the machine exhibits similar transient behaviour during starting in both cases. Rotor flux follows almost the same dynamic response and converges to its rated value, confirming that machine magnetization is not significantly affected by the form of the mechanical load.

The most significant difference concerns the evolution of speed and electromagnetic torque. For a load proportional to speed, the required torque increases linearly with speed, resulting in slower acceleration and a longer transient. In contrast, when the load is proportional to speed squared, the required torque remains lower at low speeds and increases significantly only near the rated operating region. As a result, the machine accelerates and reaches steady state faster.

In both cases, the final speed is slightly below rated speed because of the slip required to develop the load torque. For the linear load, electromagnetic torque settles at its rated value of approximately 0.77 *p.u.*, whereas for the quadratic load the corresponding value is reached only very close to rated speed.

Finally, comparison of the stator current shows that, for a load proportional to speed, the current remains around 4 *p.u.* for longer because increased electromagnetic torque is required throughout almost the entire acceleration period. In contrast, for a load proportional to speed squared, the transient is shorter and steady state is restored more quickly.

7 Scalar V/f Control

In this question, scalar V/f control was implemented for the induction motor. Its basic principle is to maintain a constant voltage-to-frequency ratio, ensuring approximately constant magnetic flux in the machine air gap.

The supply-voltage magnitude is determined by the V/f characteristic according to the following relationship. For frequencies above rated frequency, the voltage remains constant at its maximum value V_{base}.

$$|V_s| = \begin{cases} \dfrac{|\omega_{ref}|}{\omega_{base}} V_{base}, & |\omega_{ref}| \leq \omega_{base} \\ \\ V_{base}, & |\omega_{ref}| > \omega_{base} \end{cases} \tag{31}$$

Figures (19) and (20) show the implemented overall system and the scalar-controller subsystem, respectively.

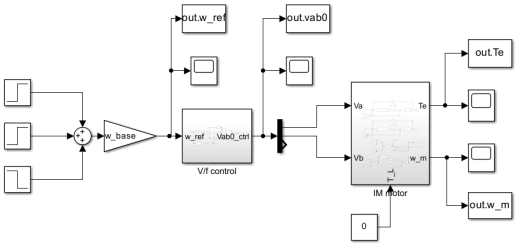


Figure 19: Base model

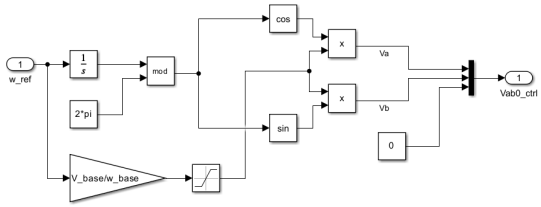


Figure 20: V/f Controller

The speed reference is defined as the sum of three step changes: first from 0 *p.u.* to 0.5 *p.u.*, then from 0.5 *p.u.* to 1 *p.u.*, and finally from 1 *p.u.* to 0.667 *p.u.*. These changes determine the corresponding motor supply frequency and voltage. Figure (21) shows the speed reference, while Figure (22) shows the voltage components in the $\alpha\beta$ frame.

Simulation Results

Figures (23)–(26) show the main machine operating quantities under scalar control and step changes in the speed reference. Load torque was kept at zero throughout the simulation.

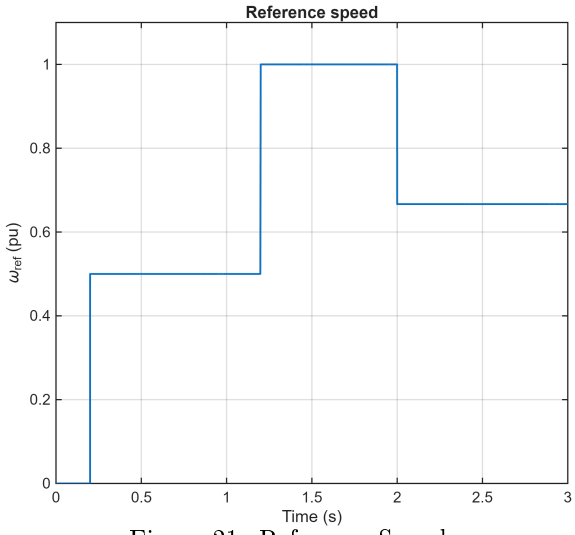


Figure 21: Reference Speed

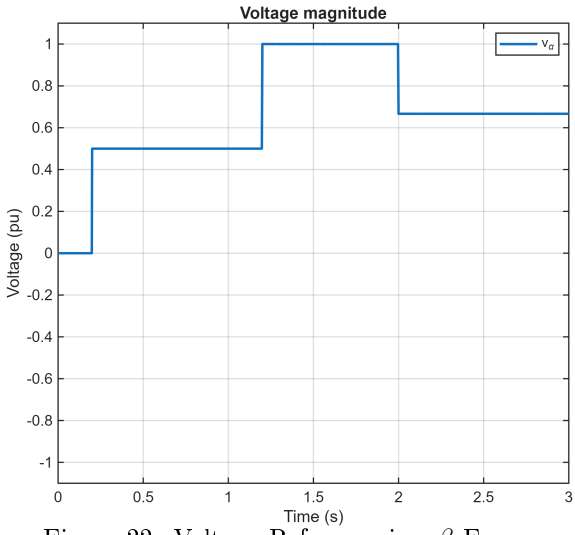


Figure 22: Voltage Reference in $\alpha\beta$ Frame

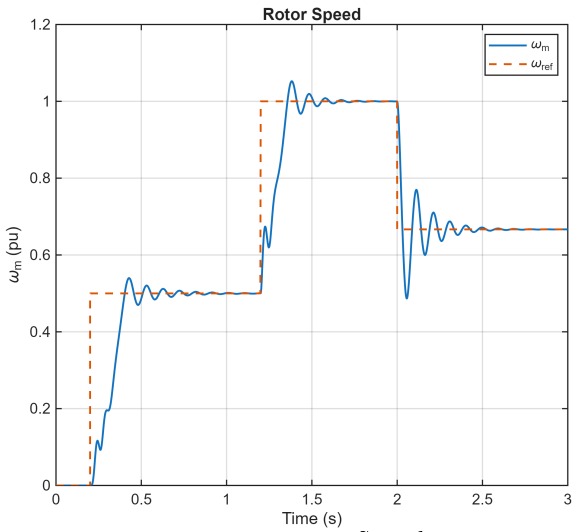


Figure 23: Rotor Speed

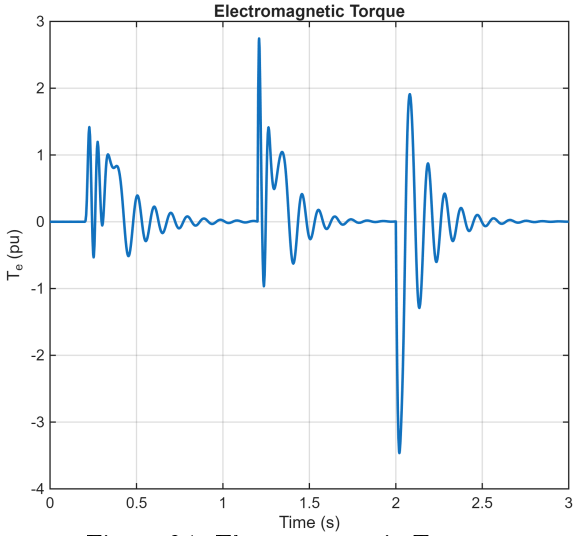


Figure 24: Electromagnetic Torque

The plots show that rotor speed follows the reference, with small oscillations during the transients.

Similarly, machine torque exhibits strong transient oscillations and remains almost zero once the machine settles at the new speed reference, since the load torque is zero.

Except during transients, rotor flux remains constant at its rated value throughout operation, confirming correct V/f control operation because the voltage-to-frequency ratio is kept constant.

Finally, stator current increases during the transient and converges to 0.4 *p.u.* in steady state because of the magnetizing current.

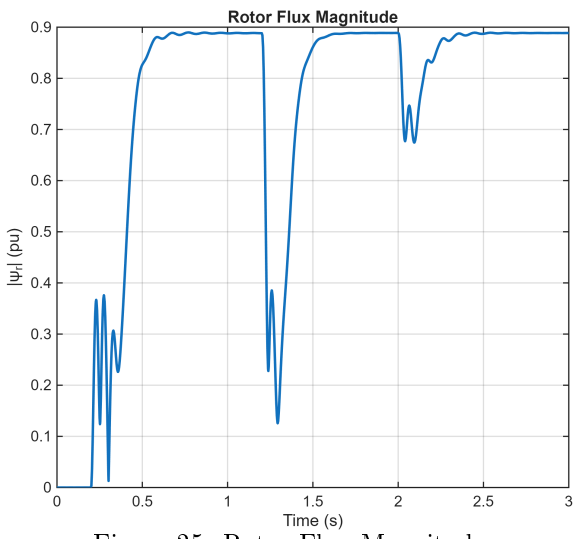


Figure 25: Rotor Flux Magnitude

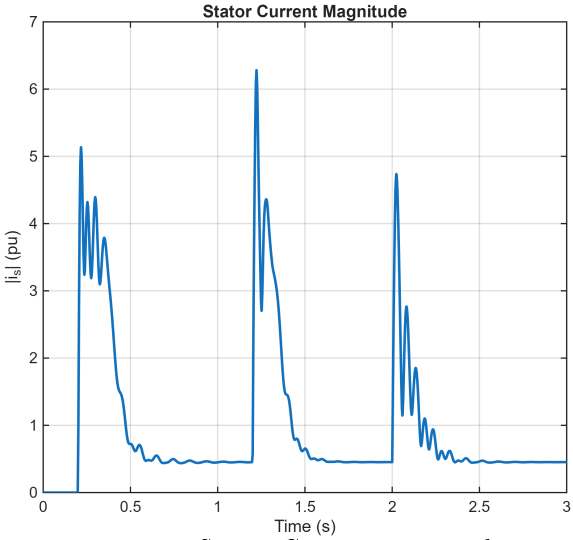


Figure 26: Stator Current Magnitude